Capacity-Valid Scheduling of a Heterogeneous GPU Cluster

2026 HKUST-GZ MCM Senior Track — Problem B

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Summary

This report analyzes the supplied heterogeneous GPU trace using one capacity-valid, non-preemptive, gang-scheduling contract. The trace contains 224,112 jobs; GPU work is concentrated (Gini 0.9522), and the largest 5% account for 87.68% of GPU work. A10 has lifetime-work/window-capacity ratio 2.064 and counterfactual immediate-start mean demand 3,614.54 GPUs against 2,494 available; these are pressure diagnostics, not measured utilization.

For Question 2, the complete 111,284-job A10 comparison gives ordinary SPT-HOL mean waiting 32.906 h versus SPT-FILL 34.967 h, against the equal-weight job-count lower bound 22.3179305700 h. A separate protected-policy experiment is limited to a verified 65,536-job prefix. For Question 3, a declared fixed-split substitution counterfactual routes a SHA-256-selected 10% subset to 60 dedicated A100 nodes with $\sigma = 1$; it reduces A10 mean waiting on this trace while retaining native-pool harm and duration sensitivity. These are finite, contract-specific observations, not optimality or hardware-performance claims.

Keywords: GPU scheduling; gang placement; heterogeneous resources; head-of-line blocking; fixed-split counterfactuals

Publication candidate. Final team identity, human review and AI disclosure require confirmation before submission.

1 Question 1: Workload Structure and Capacity Pressure

The supplied trace contains 224,112 jobs on 4,278 nodes, with 10,412 GPUs and 632,636 vCPUs across six GPU models. For job j, let a_j be release time, p_j duration in seconds, W_j worker count, and (g_j, c_j) the per-worker GPU and vCPU requests. A job's submitted GPU work is $H_j = W_j g_j p_j$. These are requests and stated durations, not measured utilization.

The data are highly concentrated: the empirical GPU-work Gini coefficient is 0.9522, the median duration is 1,330 s, the 95th percentile is 309.09 h, and the maximum is about 184 days. The top 2,241 jobs (1%) account for 43.30% of GPU work; the top 11,205 jobs (5%) account for 87.68%. The 3,512 jobs released at $a_j = 0$ are the trace-origin batch. Because no calendar origin is supplied, relative phases within a 24-hour cycle are not labelled as real-world clock periods.

For model m with capacity (G_m, C_m), the lifetime work to exact observation-window supply ratio is

$$\rho_m^{GPU} = \frac{\sum_{j \in \mathcal{J}_m} W_j g_j p_j}{T \sum_{i \in \mathcal{I}_m} G_i}, \qquad \rho_m^{CPU} = \frac{\sum_{j \in \mathcal{J}_m} W_j c_j p_j}{T \sum_{i \in \mathcal{I}_m} C_i}, \tag{1}$$

where $T = 7,775,878$ s is the exact Q1 observation window. Table 1 compares submitted lifetime work with supply; it does not claim an allocation rate or steady-state instability.

Table 1: Submitted lifetime work and exact-window capacity ratios.

GPU model	GPUs	Jobs	Work (10^6 GPU-h)	ρ^{GPU}	ρ^{CPU}
`A10`	2,494	111,284	11.119	2.064	0.391
`A100-SXM4-80GB`	3,456	80,877	5.936	0.795	0.534
`GPU-series-1`	1,558	9,643	3.892	1.156	0.590
`H800`	1,752	4,716	2.203	0.582	0.540
`GPU-series-2`	976	13,633	1.224	0.580	0.382
`A800-SXM4-80GB`	176	3,959	0.232	0.611	0.527
Total	10,412	224,112	24.606	1.094	0.473

A counterfactual immediate-start demand curve is $D_m(t) = \sum_j W_j g_j \mathbf{1}\{a_j \le t < a_j + p_j\}$. Exact interval integration gives an A10 mean of 3,614.54 GPUs against 2,494 available and an instantaneous peak of 4,103. This is a demand diagnostic under impossible immediate starts, not an observed allocation. GPU-series-1 also illustrates node-shape pressure: 2,478 jobs (25.70% of that pool, 78.07% of its work) exceed the 24-vCPU-per-GPU ratio of its 8-GPU nodes. This motivates capacity-valid scheduling tests rather than a placement conclusion from averages alone.

Q1 supplies two different lower-bound perspectives used later. The GPU-work weighted waiting bound is 560.521 h for A10 after including the drain tail; its objective weights each job by H_j. The Q2 job-count bound instead weights each job equally and is 22.3179305700 h on the complete A10 set. They are not interchangeable estimates of one common optimum. Both relax node shape, CPU coupling, and gang feasibility, so they are theoretical benchmarks rather than scheduler guarantees.

2 Question 2: Capacity-Valid Scheduling

2.1 Model and common contract

Each GPU model is simulated as a separate pool. A job starts at $s_j \geq a_j$, completes at $f_j = s_j + p_j$, and all W_j workers start simultaneously. If x_{ji} workers of job j use node i, feasibility requires

$$g_j x_{ji} \leq G_i^{free}(t), \qquad c_j x_{ji} \leq C_i^{free}(t), \qquad \sum_i x_{ji} = W_j \tag{2}$$

for every active time and selected node. The DES is non-preemptive, uses exact integer-second durations, commits a gang placement atomically, and uses first-fit by numeric node ID. All results below use the same raw-data contract, `shared-timer` event mode, $K = 200$ candidate checks per pool and timestamp, exploratory $\theta = 604800$ s, and a 90-day evaluation horizon $H = 7,776,000$ s. Waiting and JCT are computed after complete drain; the 90-day count is $f_j \leq H$.

Ordinary `SPT-HOL` orders arrived jobs by $(p_j, a_j, \mathrm{id}_j)$ and stops when the head is infeasible. `SPT-FILL` uses the same order but may inspect later feasible jobs. For the tested fill contract, $K = 200$ is a per-pool, per-timestamp bypass budget after a blocking prefix: feasible-prefix checks do not consume it, successful and failed bypass attempts do, retries may do so again, and a successful start does not reset the timestamp budget. The protected policies in the 65,536-job prefix add urgent/reservation mechanisms and are reported as a separate evidence tier; they are not relabelled as ordinary SPT-HOL.

2.2 A job-count lower bound and accounting

For the complete A10 set, let $C = 2,494$, $G_j = W_j g_j$, and $Z(t) = \{j : a_j \leq t < a_j + p_j\}$. Let $q(t)$ be the minimum number of jobs removed from $Z(t)$ so the remaining GPU width is at most C. Then

$$L_{count} = \frac{1}{3600N} \int q(t)\, dt \tag{3}$$

provides a lower bound on equal-weight mean waiting. The calculation uses exact half-open event intervals and gives 22.3179305700 h for $N = 111,284$. It is a different objective from Q1's GPU-work weighted bound.

For non-preemptive schedules, the waiting variable obeys the requested bookkeeping identity $\omega_j = \min(\omega_j, p_j) + \max(\omega_j - p_j, 0)$. We therefore report the within-window gap $[\sum_j \min(\omega_j, p_j) - \int q(t)dt]/(3600N)$ and the beyond-duration term $\sum_j \max(\omega_j - p_j, 0)/(3600N)$. This is exact bookkeeping around a capacity relaxation, not a causal allocation of blame or a removable share.

2.3 Complete A10 comparison

The complete A10 evidence has 111,284 jobs and 2,494 nodes. Independent capacity replay passed for both policies. `SPT-HOL` has mean waiting 32.906 h, P95 11.087 h, P99 1542.181 h, maximum 3017.288 h, mean JCT 132.676 h, and 107,002 jobs completed within 90 days. `SPT-FILL` has mean waiting 34.967 h, P95 15.130 h, P99 1575.462 h, maximum 3015.311 h, mean JCT 134.737 h, and 106,802 90-day completions. The paired mean difference (HOL minus FILL) is -2.060576 h: 55,266 jobs improve, 50,575 worsen, and 5,443 are unchanged;

1,226 jobs that had zero wait under FILL gain positive wait under HOL. The mean advantage is therefore trace-specific and does not establish dominance or no starvation.

Table 2: Complete-A10 results. Allocation rates use $[0, H)$; waiting/JCT use complete drain.

Policy	Mean ω	P95 ω	P99 ω	Max ω	Mean JCT	90d done
SPT-HOL	32.906	11.087	1542.181	3017.288	132.676	107,002
SPT-FILL	34.967	15.130	1575.462	3015.311	134.737	106,802

2.4 Protected-prefix evidence

The protected-policy comparison is limited to the first 65,536 A10 jobs with the same θ, K, event mode and horizon. Its count lower bound is 24.9181888580 h. The ordinary baselines have means 30.332 h (HOL) and 31.042 h (FILL); protected means are 556.492 h (urgent HOL), 350.528 h (urgent TIME), 350.088 h (urgent FIXED), and 349.715 h (urgent FLEX). FIXED and FLEX accept 62 and 133 crossing candidates and improve mean waiting relative to TIME by 0.439 and 0.813 h, while still producing both improved and worsened jobs. This is prefix evidence and cannot be extrapolated to full A10 or the full cluster.

The cross-review revision also reports GPU-work-weighted mean waits for the same complete A10 timings: 619.352 h (FCFS), 677.690 h (SPT-HOL), and 677.496 h (SPT-FILL), using $H_j = W_j g_j p_j$ as weights. Thus the weighted objective reverses the FCFS-versus-SPT ordering seen under equal-job mean waiting. The complete-A10 evidence supports ordinary SPT-HOL as the current mean-wait main candidate, with SPT-FILL retained as a useful baseline for immediate feasible starts. The protected tier remains a bounded safety experiment rather than a selected production policy.

3 Question 3: Fixed-Split Cross-GPU Substitution

3.1 Model and assumptions

We study a finite, whole-card, pre-start substitution policy on the supplied trace of 224,112 jobs. Job j has native GPU model n_j, actual execution model m_j, duration p_j, worker count W_j, per-worker GPU request g_j, and per-worker CPU request c_j. A compatibility indicator K_{jm} and multiplier σ_{jm} are declared scenario assumptions; they are not measured hardware facts. For the only foreign edge in this study, A10 jobs may execute on A100-SXM4-80GB and

$$p'_j = \sigma_{jm} p_j, \qquad f_j = s_j + p'_j, \qquad \omega_j = s_j - a_j. \tag{4}$$

The fixed partition reserves the first B A100 nodes in ascending node_name order for routed A10 jobs and leaves the other A100 nodes for native A100 jobs. Each worker's CPU and GPU remain on one physical node, all workers of a job start together, jobs are non-preemptive and non-migrating, and no resource may be used twice. At every node i and time t,

$$\sum_{j: s_j \le t < f_j} g_j y_{ji} \le G_i^{\mathrm{cap}}, \qquad \sum_{j: s_j \le t < f_j} c_j y_{ji} \le C_i^{\mathrm{cap}}, \qquad \sum_i y_{ji} = W_j. \tag{5}$$

The model is a deterministic scheduling heuristic, not an optimality claim.

3.2 Fixed-split counterfactual design

M0 is the original model-isolated schedule. S0 is the earlier Q3-S1 M1 result, renamed to make the comparison unambiguous: 60 dedicated A100 nodes, a 10% SHA-256 threshold, and $\sigma = 1$. S1–S5 vary exactly one factor from S0: 5% or 20% routing, 30 or 90 donor nodes, or $\sigma = 2$. The hash namespace is fixed as 'q3-fixed-split-v1'; eligibility uses only the current job and declared capacity. There is no fallback from the foreign queue and no relocation of a running job.

We report worker count W_j separately from waiting $\omega_j = s_j - a_j$, actual JCT $J_j = f_j - a_j$, relative completion delay $f_j - a_j - p_j$, and paired finish difference $f_{\text{scenario}} - f_{\text{reference}}$. Every comparison uses the same original job IDs. The 90-day window is $H = 7,776,000$ seconds; waiting and JCT use complete drain, while completion counts use $f_j \le H$.

Table 3: All fourteen formal schedules: waiting, tail and native-harm rows from the r4 paper-ready metrics.

Scenario	Policy	A10 $\bar{\omega}$ (h)	P95/P99 (h)	Native harm	Routed%
M0	FCFS	900.129	1408.478/1419.616	0	0%
S0	FCFS	587.875	966.761/987.569	0	10%
S1	FCFS	702.551	1167.986/1188.658	0	5%
S2	FCFS	1010.136	4322.906/5345.115	0	20%
S3	FCFS	916.407	4053.782/4423.899	0	10%
S4	FCFS	539.109	958.246/985.841	5,735	10%
S5	FCFS	721.752	2292.733/2885.707	0	10%
M0	SJF-HOL	32.906	11.087/1542.181	0	0%
S0	SJF-HOL	20.534	12.691/639.532	0	10%
S1	SJF-HOL	25.328	7.745/1108.877	0	5%
S2	SJF-HOL	22.058	9.767/526.165	0	20%
S3	SJF-HOL	25.565	11.470/1025.261	0	10%
S4	SJF-HOL	19.214	5.205/611.002	2,150	10%
S5	SJF-HOL	25.106	12.125/1058.270	0	10%

3.3 Benefit, native-pool cost and sensitivity

With $\sigma = 1$, S0 reduces A10 mean waiting from 900.129 to 587.875 hours under FCFS (34.7% relative to M0) and from 32.906 to 20.534 hours under SJF-HOL (37.6%). Cluster-wide mean JCT changes from 523.961 to 368.910 hours for FCFS and from 93.353 to 87.209 hours for SJF-HOL. The native A100 mean and P99 waiting costs are zero in S0 on this trace; this is a trace result, not an SLA.

Under S0, SJF-HOL versus M0 makes 49,772 A10 jobs finish later, gives 4,397 jobs that had zero M0 waiting a new delay, and has 68183.1825 positive completion-delay job-hours. Its A10 waiting P95 changes from 11.086875 to 12.690806 hours. These are S0-minus-M0 paired quantities; the harmed-job denominator is not used for the all-job P95.

For the 90-node S4 point, the native denominator is separated explicitly. FCFS has 80,877 native A100 jobs in the full population, of which 5,735 have positive S4-minus-S0 waiting;

Table 4: The same fourteen schedules: JCT and 90-day completion counts. Waiting and JCT use complete drain; completion counts use the 90-day window.

Scenario	Policy	A10 $\bar{J}$ (h)	All $\bar{J}$ (h)	90-day done
M0	FCFS	999.898	523.961	165,107
S0	FCFS	687.645	368.910	181,112
S1	FCFS	802.320	425.853	174,693
S2	FCFS	1109.905	578.586	177,498
S3	FCFS	1016.177	532.044	174,139
S4	FCFS	638.878	344.723	184,339
S5	FCFS	831.456	440.320	177,166
M0	SJF-HOL	132.676	93.353	217,570
S0	SJF-HOL	120.303	87.209	217,814
S1	SJF-HOL	125.097	89.589	217,766
S2	SJF-HOL	121.827	87.965	218,104
S3	SJF-HOL	125.334	89.707	217,762
S4	SJF-HOL	118.984	86.560	217,945
S5	SJF-HOL	134.810	94.412	217,521

the corresponding SJF-HOL counts are 80,877 and 2,150. The all-native mean additional waits are 0.077111 and 0.018492 hours (FCFS/SJF-HOL), whereas the harmed-subgroup means are 1.087453 and 0.695606 hours. Positive additional-wait totals are 6236.5408 and 1495.5528 job-hours. The all-native P99 values are 2.074778 and 0.405622 hours; harmed-subgroup P99 values, computed only from positive-delta jobs, are 4.099278 and 6.072953 hours. Every value in this paragraph is S4 minus S0, with its population stated.

More routing is not uniformly better. At 60 donor nodes, FCFS A10 mean waiting is 702.551, 587.875, and 1010.136 hours at 5%, 10%, and 20%. The 20% case is worse than both 10% and M0. SJF-HOL gives 25.328, 20.534, and 22.058 hours, while its P99 tail and 90-day completion count move differently. These are finite fixed-split observations; they do not establish monotonicity for another trace or a dynamic policy.

At 10% routing and $\sigma = 1$, increasing donors from 30 to 60 to 90 lowers A10 mean waiting for both policies, but 90 nodes creates the native costs reported above. A compact tail check retains the negative cases: S2 FCFS (20% routing, 60 donors) reaches A10 maximum waiting 5445.367 hours, while S3 FCFS (10%, 30 donors) has mean waiting 916.407 hours, worse than M0's 900.129 hours.

For $\sigma = 2$, the effective duration of routed A10 jobs doubles and SJF-HOL is re-simulated because its queue key changes. FCFS A10 mean JCT changes from 999.898 to 831.456 hours (paired delta -168.443 hours), while SJF-HOL changes from 132.676 to 134.810 hours (paired delta 2.134 hours). For the full trace, SJF-HOL all-job mean JCT changes from 93.353 to 94.412 hours and 90-day completions from 217,570 to 217,521; waiting falls relative to M0 but completion worsens. FCFS retains a mean-JCT benefit in this $\sigma = 2$ scenario. No interpolation or exact break-even σ^* is claimed.

3.4 Conditional recommendation and limitations

If the policy requirement is zero additional native A100 waiting on this trace, 60 nodes at 10% is a tested candidate for either policy; it is not a proven maximum safe capacity or a global

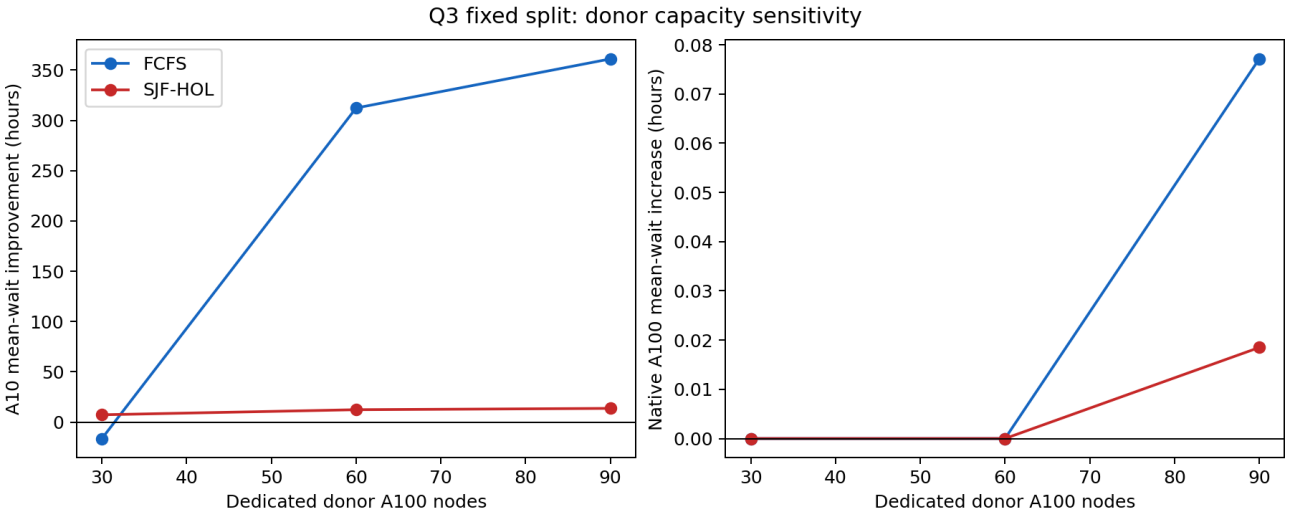


Figure 1: Capacity sensitivity from the persisted S2 results. The left panel shows A10 mean-wait improvement relative to M0; the right panel shows native A100 mean-wait increase relative to M0. The two panels use different scales so the small native cost remains visible without being treated as zero.

optimum. If a small native cost is acceptable, 90 nodes supplies an additional point on the A10–native trade-off. SJF-HOL's lower mean waiting does not remove its tail and completion-fairness losses; all harmed-job counts and positive delay totals remain in the paired evidence.

Compatibility and σ are assumptions without an A10/A100 performance matrix. Allocation rates are request occupancy, not measured utilization; GPU-hours across models are not interchangeable compute or monetary units. The experiment adds no physical GPUs, does not implement untested cross-model mechanisms, migration, or Q2 $\times$ Q3 scheduling, and does not prove heuristic optimality. Q1's A10 lower bound is not transferred to substituted schedules. The conclusion is therefore conditional on the declared trace, partitions, hash rule, policy, and duration scenarios.

4 Conclusion and evidence limits

The three questions provide complementary evidence under a declared trace and simulation contract. Q1 identifies workload concentration and two lower bounds with different weighting objectives. Q2 compares ordinary policies on the complete A10 trace and keeps protected policies at their verified prefix scope. Q3 evaluates a fixed, physical, pre-start substitution scenario while retaining native-pool harm and duration sensitivity.

Q1 uses the exact observation window $T = 7,775,878$ s; scheduling waiting and JCT use complete drain; 90-day completion counts use $H = 7,776,000$ s. Prefix evidence is not extended to the complete A10 trace. No unsupported performance, hardware-speed, pricing, or optimality claim is used. Large Q3 delivery archives remain local with hashes; cross-machine availability is unresolved.

References

1. HKUST-GZ MCM 2026 Senior Track, Problem B, repository copy: `problems/2026_gpu_scheduling/docs/2026_MCM_GPU_Scheduling.pdf`; SHA256 is recorded in

`paper_sources.json`.

2. Q1 verified evidence, fixed commit `7075317883808b1cd7f80cadf3980be291c69cf0`; paths and hashes are in `paper_sources.json`.
3. Q2 cross-review evidence, fixed commit `b91918bb084848b93de642ad4e3ea0a7d3f25e68`; paths and hashes are in `paper_sources.json`.
4. Q3 cross-review publication evidence, fixed commit `170cc37a820e830a969cb1b1cd3cadb1e8fdf256`; r4 path `results/20261016-b-q3-s3-publication-9ac1d7f2-r4/`; r3 is historical.
5. Claude independent archive, fixed commit `9aeddd708b29` (full SHA in `paper_sources.json`); preserved for provenance and review, not executed in this release.
6. Submission portal: https://cmt3.research.microsoft.com/HKUSTGZMCM2026. The repository does not contain the separate Registration and Submission Guidelines PDF; its availability is recorded as BLOCKED pending team confirmation.

Report on use of AI tools

This integration used OpenAI Codex to inspect fixed commits, combine already saved Q1/Q2/Q3 evidence, revise and typeset this candidate, and run repository and PDF checks. The Claude package was preserved as an independent source; its scripts were not executed here. The team must confirm whether ChatGPT, Claude, or other tools were used outside this record and must approve the final wording. No AI output is treated as experimental evidence without a fixed source and human review.

Submission status: SUBMISSION_BLOCKED pending team names, institution/advisor fields, human review, and confirmed AI-use record.